LAWS OF MOTION

Newtonian Mechanics
1642 - 1727

Force is that cause which changes the state of the body.

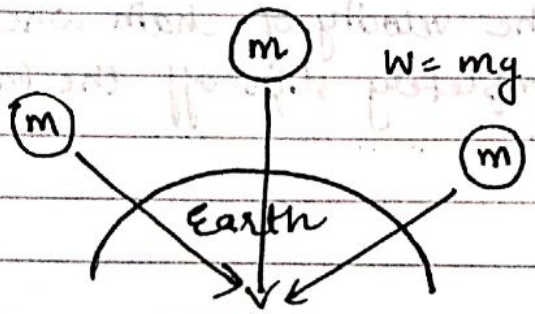
SI unit ⟹ Newton
CGS unit ⟹ Dyne

Force is a vector quantity.

Non-contact Force	Contact Force (Electromagnetic in nature)	Weak Force	Nuclear force
• Gravitational force	• Tension (T)	Van der waal forces	
• Electrostatic force	• Normal Reaction (R)		
• Magnetic force	• Spring force (F_s / F_{sp})		
	• Friction		

GRAVITATIONAL FORCE

Attraction between objects due to virtue of their masses.
(Weight)

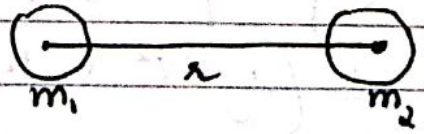


$W = mg$

acts towards the centre of Earth

1kgf = 10N / 9.8N

$$f = \frac{Gm_1m_2}{r^2}$$

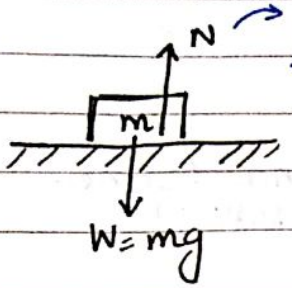


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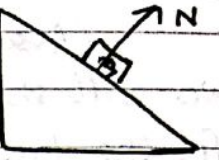
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NORMAL REACTION (N)



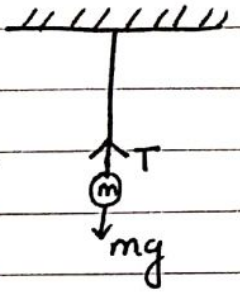
→ ⊥ to surface specifically perpendicular to point of contact



Normal reaction is caused due to contact and balaces the weight of the body.

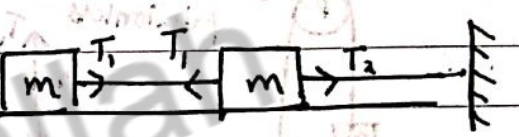
TENSION (T)

String / Rod / chain

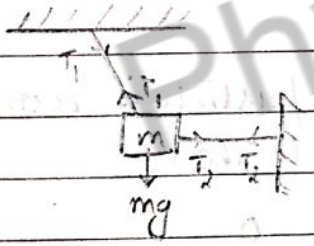


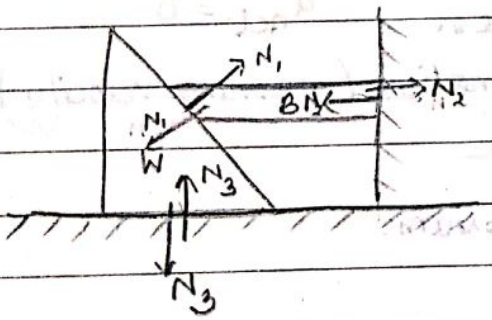
Direction of tension is away from point

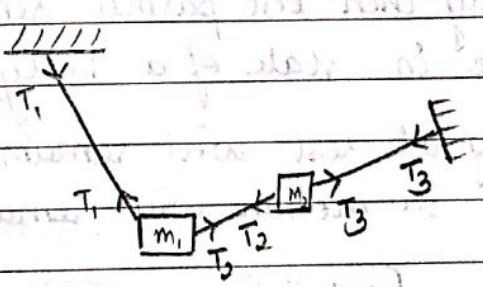
String change leads to tension change.



Tension remains same until string remains same as long as string is massless.







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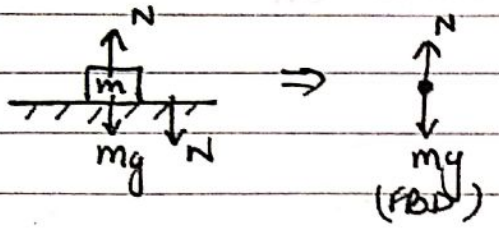
Spiral

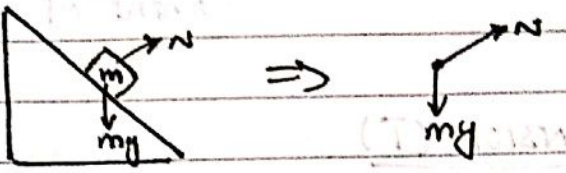
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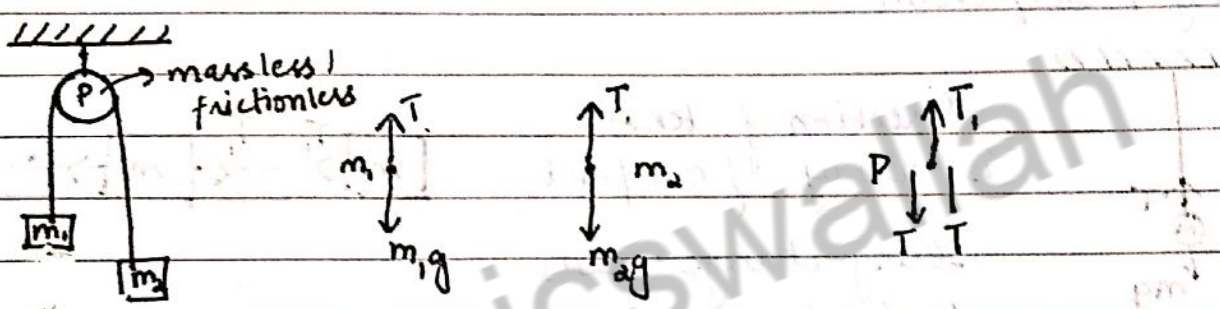
FREE BODY DIAGRAM (F.B.D.)

- Body is shown as point mass. (Assumption)
- We will show all forces acting on it.
- For the force, we should have a reference frame. (observer)

Ground ⇒ Inertial frame

(i) 

(ii) 

(iii) 

NEWTON'S FIRST LAW OF MOTION

If the vector sum of all the forces acting on a particle is zero then and only then the particle remains unaccelerated.

(No change in state of a body) $F_{net} = 0$ $a_{net} = 0$

A body at rest will remain at rest

A body in motion will remain in motion (constant velocity)

$F_{net} = 0$ ⌊ called Equilibrium.

$F_x = 0$

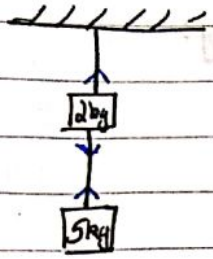
$F_y = 0$

$F_z = 0$

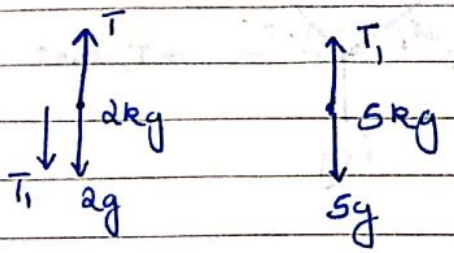
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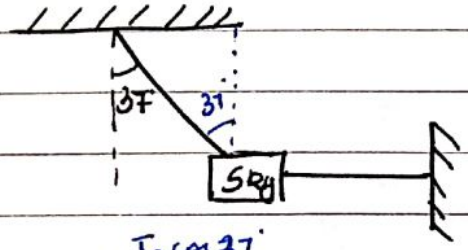
Q. 

Find Tension in both strings.
The whole system is in equilibrium.

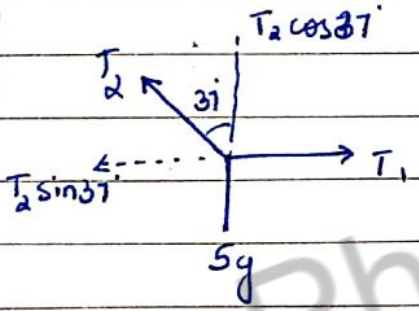


As system is in equilibrium,
$T_1 = 5g$
$\Rightarrow T_1 = 50\,N$

&
$T = 2g + T_1$
$\Rightarrow T = 20 + 50 = 70\,N$

Q. 

Find tension in both strings. (The whole system is in eqm)



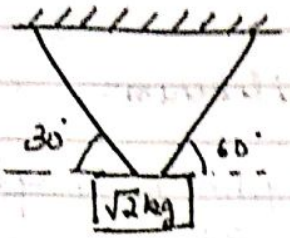
$\therefore T_2 \cos 37^\circ = 5g$
$\Rightarrow T_2 \frac{4}{5} = 50$
$\Rightarrow T_2 = \frac{250}{4} = \underline{62.5\,N.}$

&
$T_1 = T_2 \sin 37$
$\Rightarrow T_1 = \frac{50}{4} \times \frac{3}{5}$
$\Rightarrow T_1 = 37.5\,N$

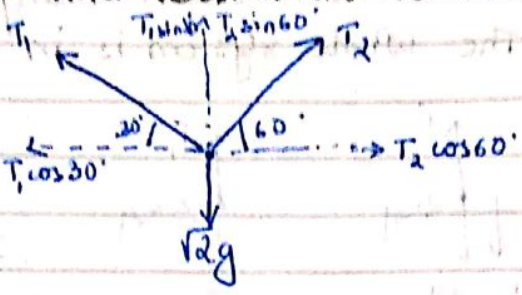
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Q



Find tension in both strings.



$T_1 \cos 30^\circ = T_2 \cos 60^\circ$

$\Rightarrow T_1 \frac{\sqrt{3}}{2} = \frac{T_2}{2} \times 1 \quad \Rightarrow T_2 = \sqrt{3} T_1$

$T_1 \sin 30^\circ + T_2 \sin 60^\circ = \sqrt{2} g$

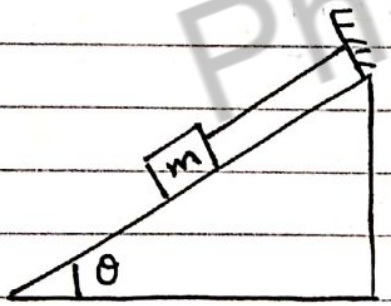
$\Rightarrow T_1 \times \frac{1}{2} + T_2 \frac{\sqrt{3}}{2} = \sqrt{2} \times 10$

$\Rightarrow T_1 + \sqrt{3} T_2 = 20\sqrt{2}$

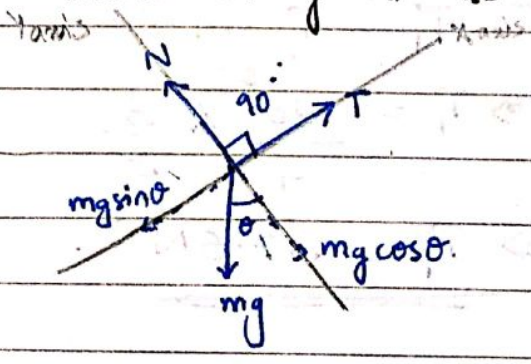
$\Rightarrow T_1 + 3T_1 = 20\sqrt{2} \quad \Rightarrow T_1 = 5\sqrt{2}\ N$

$T_2 = 5\sqrt{6}\ N$

Q



Find T in string & N Reaction on mass m if 'm' is in equilibrium.



$N = mg\cos\theta$

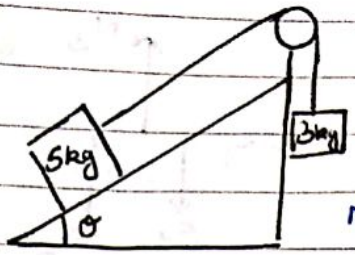
$T = mg\sin\theta$

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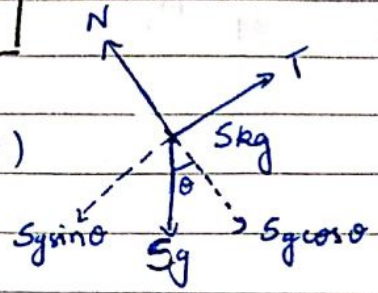
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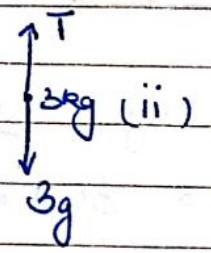
Q



If the system is in equilibrium, find θ, T and N.



(i)



(ii)

$T = 3g = 30\ N$. [from (ii)]

Also, $T = 5g \sin\theta$

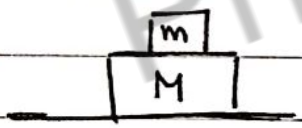
$\Rightarrow 30 = 50 \sin\theta$

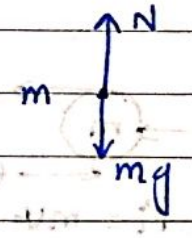
$\Rightarrow \sin\theta = \frac{3}{5}$ $\quad \therefore \theta = 37°$

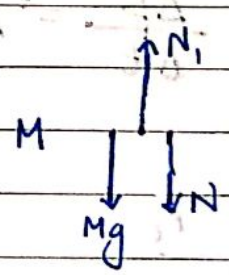
$N = 50 \times \frac{4}{5} = 40\ N$

Q

Find N between two blocks and N between surface and mass M.







$N = mg$

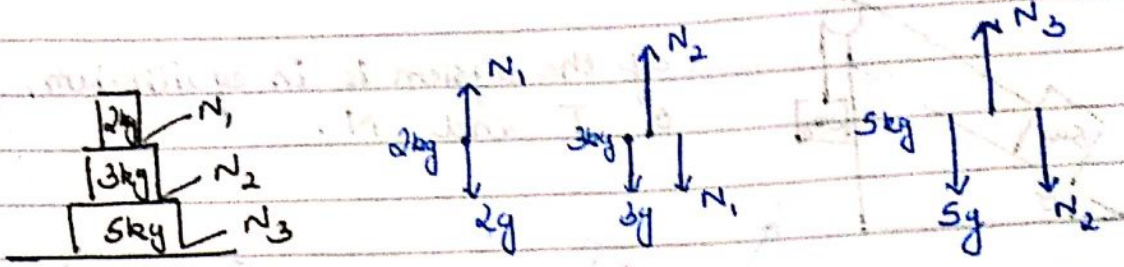
$N_1 = Mg + N$

$\Rightarrow N_1 = Mg + mg$

$\Rightarrow N_1 = (M+m)g$

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$N_1 = 2g = \underline{20\,N}$.

$N_2 = 30 + 20 = \underline{50N}$

$N_3 = 50 + 50 = \underline{100N}$

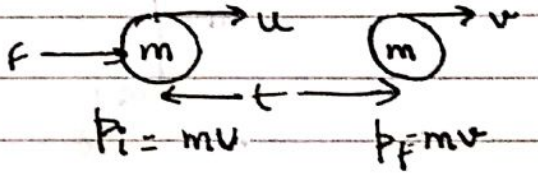
NEWTON'S SECOND LAW OF MOTION

The acceleration of a particle as measured from an inertial frame is given by the vector sum of all forces acting on the particle divided by its mass.

Rate of change in momentum = force applied

$$\frac{\Delta p}{\Delta t} = \text{force}$$

We know that,

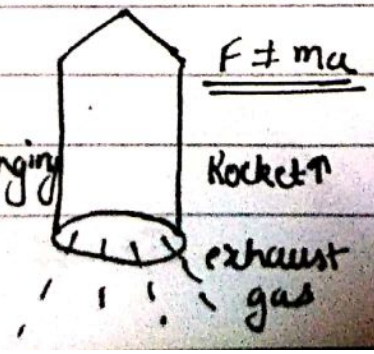


$$F = \frac{\Delta P}{\Delta t} = \frac{P_f - P_i}{t}$$

$$F = \frac{mv - mu}{t}$$

$$F = m\left(\frac{v-u}{t}\right)$$

$F = ma$] true as long as mass is constant



$F \neq ma$

$F = \frac{\Delta p}{\Delta t}$ Always valid.

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#CASE

If Object speed $v \approx c$ (speed of light)
mass changes (increases)

$$\text{Actual mass } m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$ ← Rest mass

eg when $v = \frac{c}{2}$, $m = \frac{2m}{\sqrt{3}}$

DIFFERENTIAL FORM OF NEWTON'S SECOND LAW

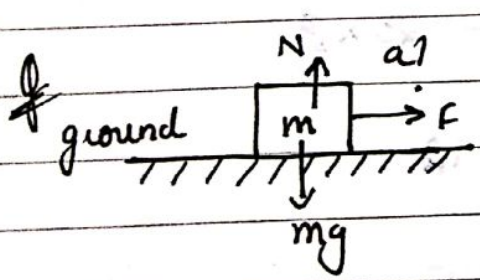
When mass is variable,

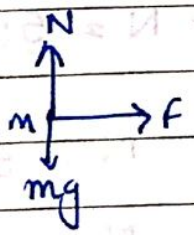
$$F = \frac{\Delta p}{\Delta t} = \frac{dp}{dt}$$

$$F = \frac{d(mv)}{dt} = m\frac{dv}{dt} + v\frac{dm}{dt}$$

$$\boxed{F = m\frac{dv}{dt} + v\frac{dm}{dt}}$$ ⇒ Exact 2nd law of motion.

- $m\frac{dv}{dt}$: cause of force when velocity changes
- $v\frac{dm}{dt}$: cause of force when mass changes.





$N = mg$

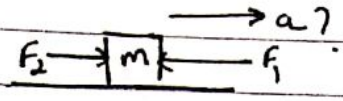
$F = ma$

$\Rightarrow a = \frac{F}{m}$

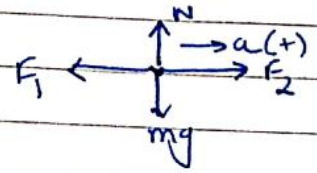
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Q



$F_2 > F_1$

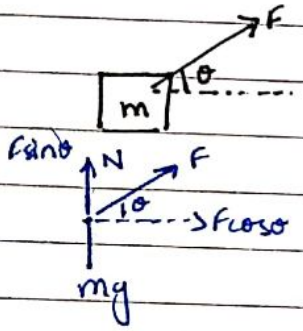


$N = mg$

$F_2 - F_1 = ma$

$\Rightarrow a = \frac{F_2 - F_1}{m}$

Q



Find N and a of body.

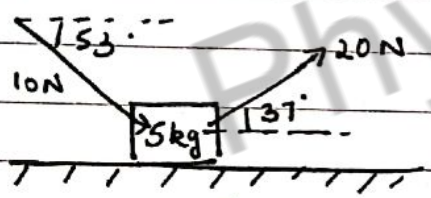
$N + F\sin\theta = mg$

$\Rightarrow N = mg - F\sin\theta$

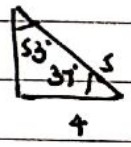
$F\cos\theta = ma$

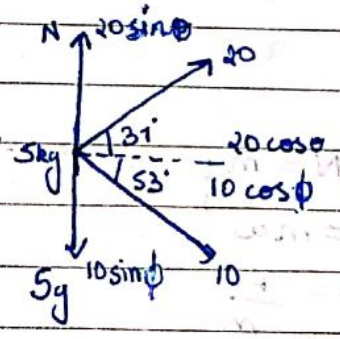
$\Rightarrow a = \frac{F\cos\theta}{m}$

Q



Find N and a of body.





$N + 20\sin\theta = 50 + 10\sin\phi$

$\Rightarrow N = 50 + \frac{10 \times 4}{5} - \frac{20 \times 3}{5}$

$\Rightarrow N = 50 + 8 - 12$

$\Rightarrow N = 46\ N$

$20\cos\theta + 10\cos\phi = 5 \times a$

$\Rightarrow 5a = \frac{20 \times 4}{5} + \frac{10 \times 3}{5} = 16 + 6$

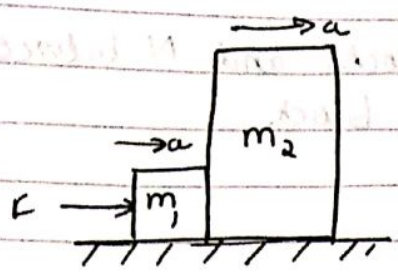
$\Rightarrow a = \frac{22}{5} = 4.4\ ms^{-2}$

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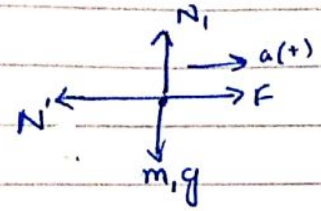
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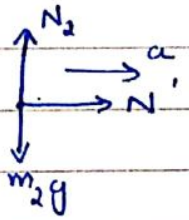


Find N between blocks & a of both blocks.



$N_1 = m_1 g$

$F - N' = m_1 a$ — (i)



$N_2 = m_2 g$

$N' = m_2 a$ — (ii)

Adding eqn (i) & (ii).

$F = (m_1 + m_2) a$

$\Rightarrow a = \frac{F}{(m_1 + m_2)}$

$N' = \frac{F m_2}{m_1 + m_2}$

Shortcut: Considering both bodies a system.

$M_{net} = m_1 + m_2$

$F_{net} = m_{net} a$

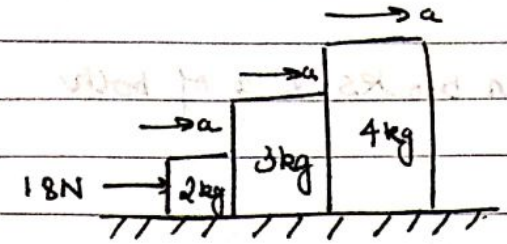
$\therefore a = \frac{F}{m_1 + m_2}$

Physics Wallah

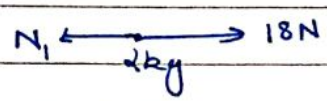
Spiral

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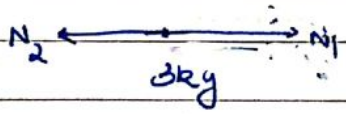
Q. 

Find a of all blocks and N between 4kg and 3kg block

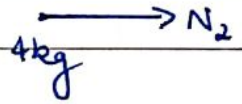


$18 - N_1 = ma$

$\Rightarrow 18 - N_1 = 2a$ —(i)



$N_1 - N_2 = 3a$ —(ii)



$N_2 = 4a$ —(iii)

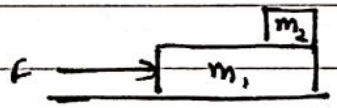
Adding all 3 eqⁿ,

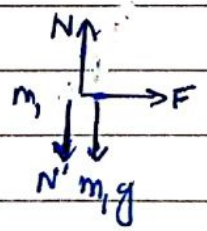
$18 = 9a$

$\therefore a = 2\ ms^{-2}$

$N_2 = 4 \times 2 = 8N$

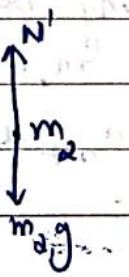
Q. Find a of m_1 & m_2.
[All surfaces are smooth]





$F = m_1 a$

$\therefore a_1 = \frac{F}{m_1}$



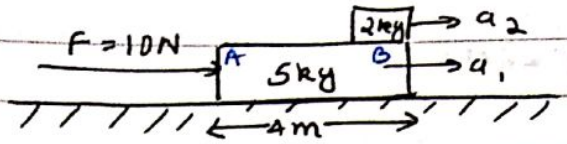
$a_2 = 0$

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Q



All surfaces are smooth. Find the time in which 2kg falls off 5kg.

$a_1 = \frac{10}{5} = 2 ms^{-1}$

$a_2 = 0$

$a_{BA} = a_B - a_A$

$= 0 - 2 = -2$

$s_{BA} = -4m$

$\mu_B = 0$

$t = ?$

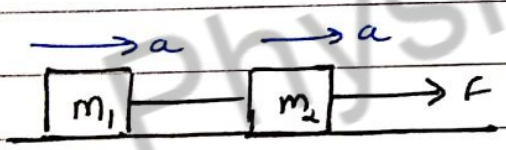
$s = ut + \frac{1}{2}at^2$

$\Rightarrow -4 = 0 + \frac{1}{2} \times -2 \times t^2$

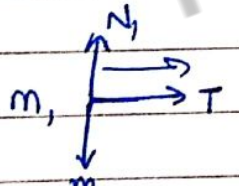
$\Rightarrow t = \pm 2$

$t = 2s$

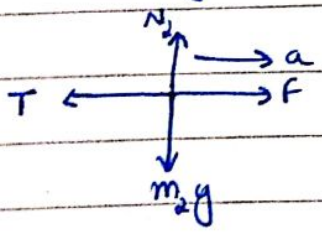
Q



Find T in string.
Find a of m_1 & m_2



$T = m_1 a$ —(i)



$F - T = m_2 a$ —(ii)

Adding both eqn,

$F = m_1 a + m_2 a$

$\Rightarrow F = a(m_1 + m_2)$

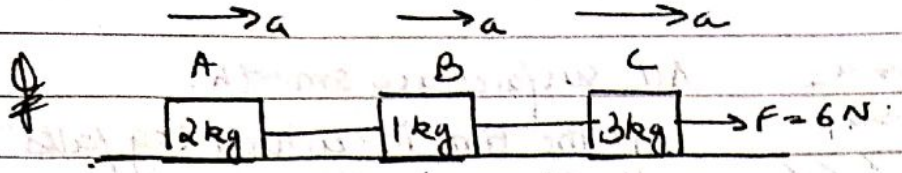
$\Rightarrow a = \frac{F}{m_1 + m_2}$

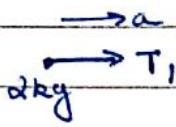
$T = \frac{F m_1}{m_1 + m_2}$

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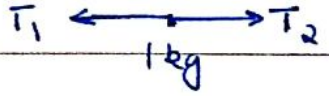
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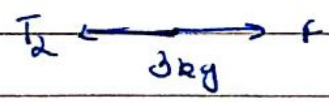




$T_1 = 2a$ — (i)



$T_2 - T_1 = a$ — (ii)



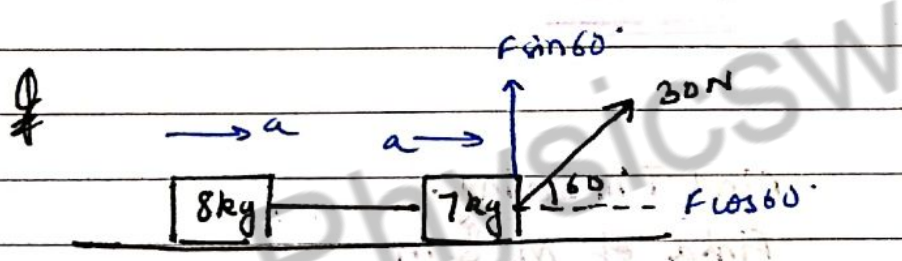
$F - T_2 = 3a$ — (iii)

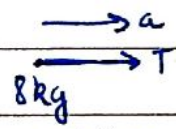
Adding 3 eq^n,

$\Rightarrow F = 6a$

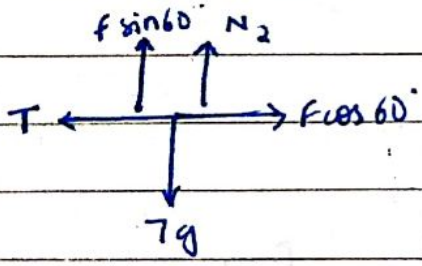
$\Rightarrow 6 = 6a$

$\Rightarrow a = 1\ ms^{-2}$





$T = 8a$ — (1)



$F\cos 60° - T = 7a$

$\Rightarrow 30 \times \frac{1}{2} - T = 7a$

$\Rightarrow 15 - T = 7a$ — (2)

Adding both eq^n,

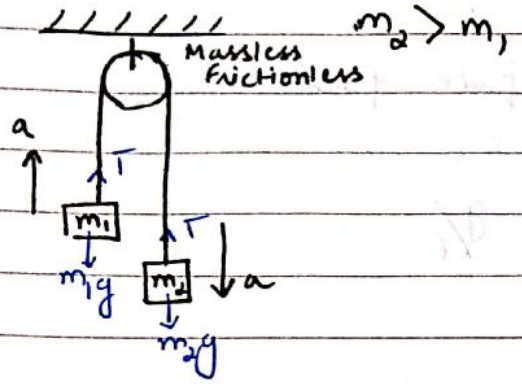
$15 = 15a$

$\Rightarrow a = 1\ ms^{-2}$

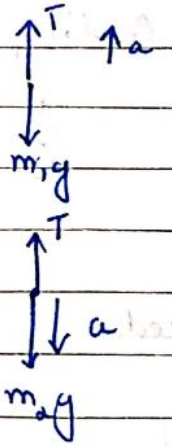
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PULLEY SYSTEM / ATWOOD MACHINE



$m_2 > m_1$



$$T - m_1 g = m_1 a \quad \text{—(i)}$$

$$m_2 g - T = m_2 a \quad \text{—(ii)}$$

Adding both equations,

$$m_2 g - m_1 g = m_1 a + m_2 a$$

$$\therefore \boxed{a = \frac{(m_2 - m_1)\, g}{m_1 + m_2}}$$

System

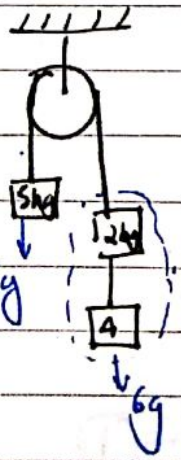


Net force $= m_2 g - m_1 g$

Net mass $= m_1 + m_2$

$$a = \frac{(m_2 - m_1)\, g}{m_1 + m_2}$$

Q



Find acceleration.

$$a = \frac{F_{net}}{M_{net}}$$

$$= \frac{6g - 5g}{11}$$

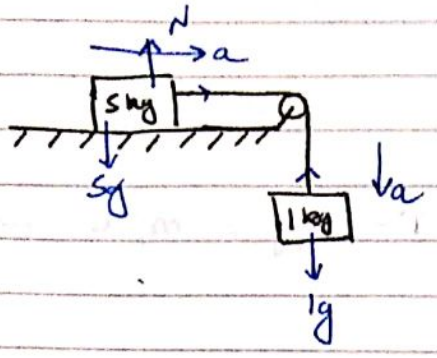
$$= \frac{10}{11}\ ms^{-2}$$

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Q



$g - T = a$

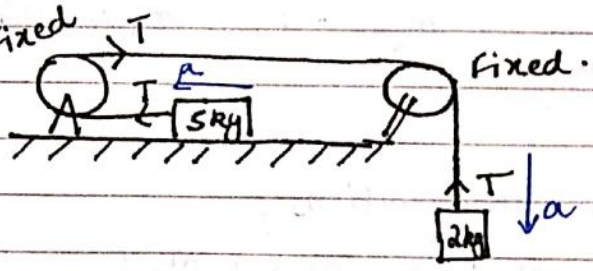
$T = 5a$

Adding both equations,

$g = 6a$

$a = g/6$

Q



If the direction of motion is upwards, acceleration is (+ve) else downwards is (-ve)

$2g - T = 2a$

$T = 5a$

Adding both equations,

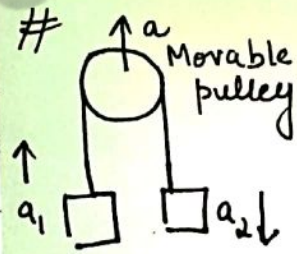
$2g = 7a$

$\Rightarrow a = \frac{2g}{7}$

or

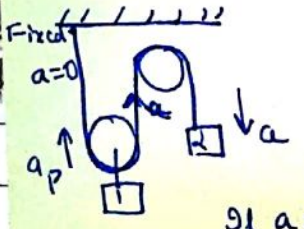
$a = \frac{2g}{7}$

#



$a = \frac{a_1 + a_2}{2}$ (with sign)

$\hookrightarrow a = \frac{a_1 - a_2}{2}$

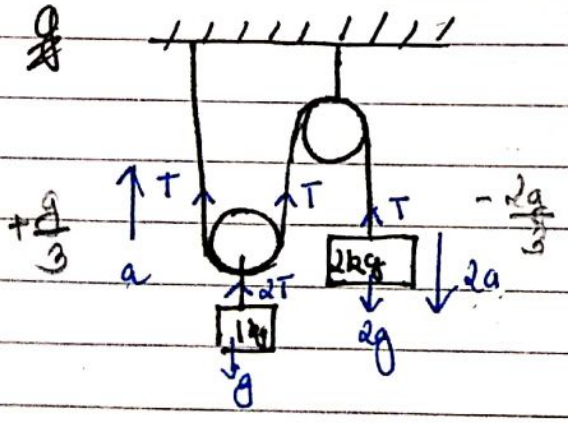


$a_p = \frac{a+0}{2}$

$\Rightarrow a_p = \frac{a}{2}$

If $a = 2a$ then $a_p = a$

Q



$2g - T = 2 \times 2a = 4a$ ⇒

$2T - g = a$ — (2)

Adding both eqn

$3g = 9a$

$\Rightarrow a = g/3$

Spiral

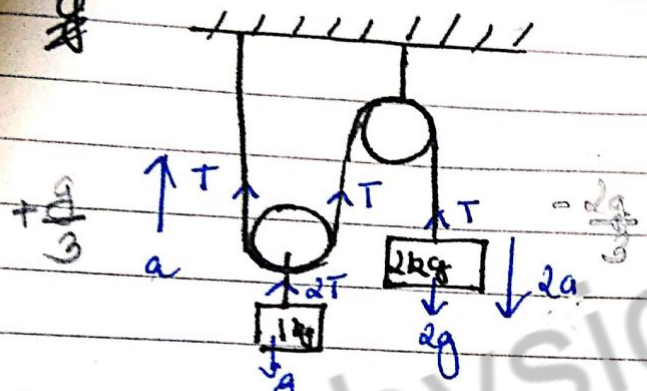
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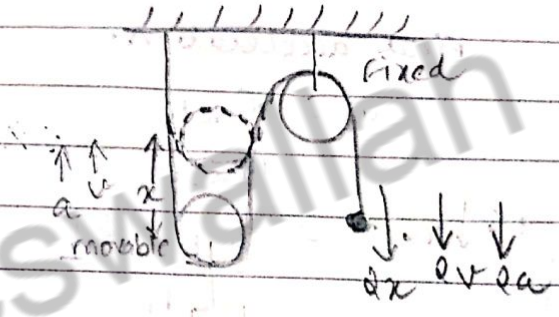
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Adding both equations,

$2g = 7a$

$\Rightarrow a = \frac{2g}{7}$





$2g - T = 2 \times 2a = 4a \Rightarrow 4g - 2T = 8a$ — (1)

$2T - g = a$ — (2)

Adding both eqⁿ

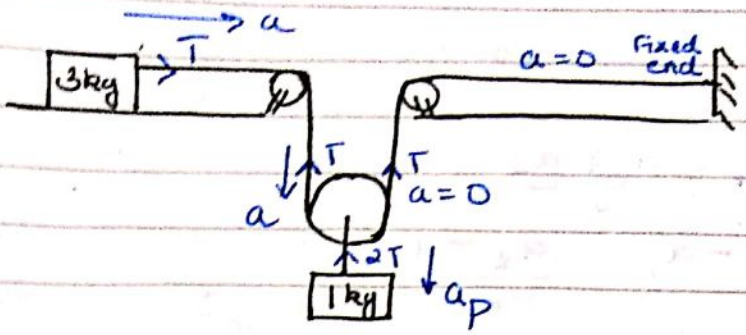
$3g = 9a$

$\Rightarrow a = g/3$

Spiral

Physics Gallah

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$$a_p = \frac{a_1 + a_2}{2}$$

$$\Rightarrow -a_p = \frac{-a + 0}{2}$$

$$\therefore a_p = \frac{a}{2}$$

Let the acceleration of 3kg block be $2a$.

Acceleration of pulley $= \frac{a}{2}$.

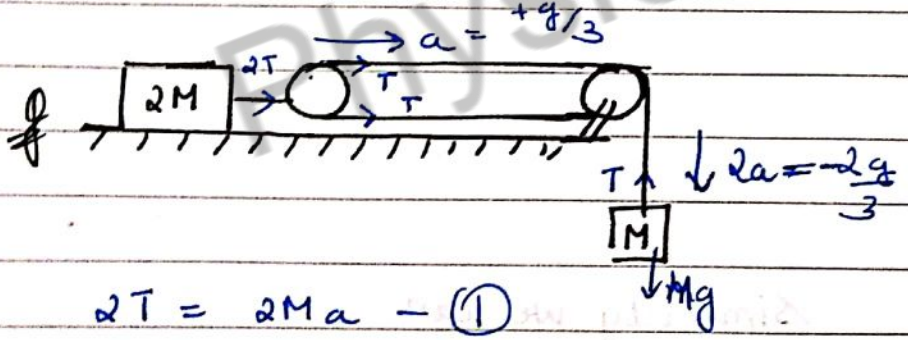
$$1g - 2T = a \quad -(1)$$

$$T = 6a \Rightarrow 2T = 12a \quad -(2)$$

Adding both equations,

$$g = 13a$$

$$\therefore a = \frac{g}{13}$$



$$2T = 2Ma \quad -(1)$$

$$Mg - T = 2Ma$$

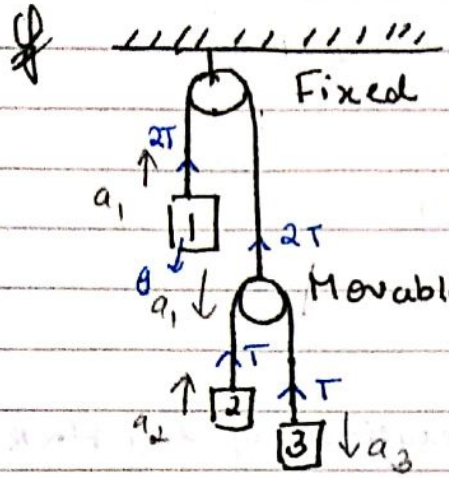
$$\Rightarrow 2Mg - 2T = 4Ma \quad -(2)$$

$$2Mg = 6Ma$$

$$\Rightarrow a = \frac{g}{3}$$

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$$a_p = \frac{a_1 + a_2}{2}$$

$$\Rightarrow -a_1 = \frac{a_2 - a_3}{2}$$

$$\Rightarrow -2a_1 = a_2 - a_3 \quad \text{---} (1)$$

$$\left.\begin{aligned} 2T - g &= a_1 \\ 3g - T &= 3a_3 \\ T - 2g &= 2a_2 \end{aligned}\right\} (2)$$

Taking eqn (1),

$$-2(2T - g) = \frac{T - 2g}{2} - \frac{3g - T}{3}$$

$$\Rightarrow -4T + 2g = \frac{3T - 6g - 6g + 2T}{6}$$

$$\Rightarrow -24T + 12g = 5T - 12g$$

$$\Rightarrow 24g = 29T$$

$$\Rightarrow T = \frac{24}{29}g$$

$$2 \times \frac{24}{29}g - g = a_1$$

$$\Rightarrow \frac{48g - 29g}{29} = a_1$$

$$\Rightarrow \frac{19}{29}g = a_1$$

Similarly we can find a_2 & a_3

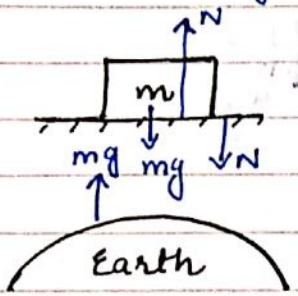
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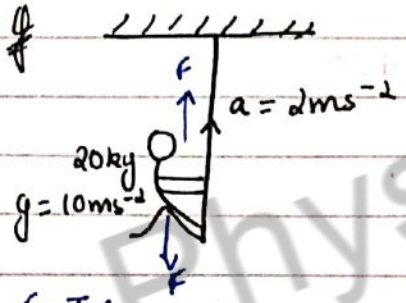
NEWTON'S THIRD LAW OF MOTION

Every action has an equal and opposite reaction.
Action - reaction pair.

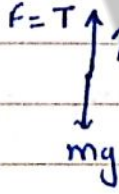
① Action & reaction acts on different bodies.
② Action - reaction pair are of same nature.
③ Force always exist in pairs.



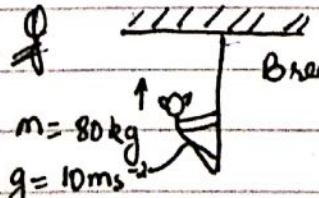
N and mg are not action - reaction pair.
mg on block is paired with mg on earth.
N on block is paired with N on surface.



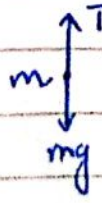
Tension in string,



$T - mg = ma$
$T = ma + mg$
$T = 20 \times 10 + 20 \times 2$
$= 240\ N$



With what maximum acceleration can the monkey safely climb the rope?



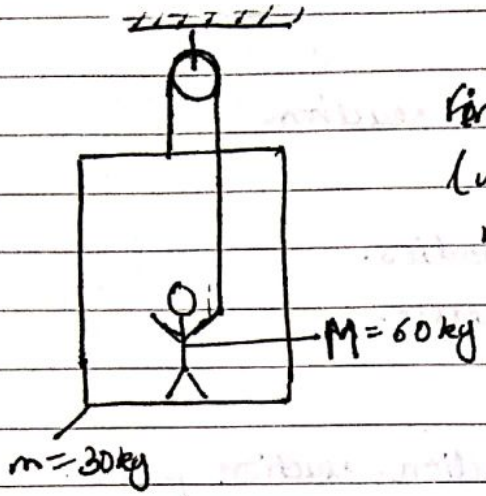
$T - mg = ma \Rightarrow T = ma + mg$.
$T \leq 1000\ N$ so $T_{max} = 1000\ N$
$\Rightarrow 80(a+10) = 1000\ N$
$\therefore a = \frac{1000}{80} - 10 = \frac{20}{8} = 2.5\ ms^{-2}$

Spiral

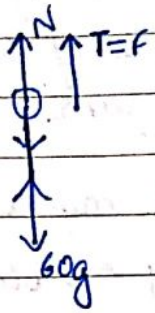
Physicswallah

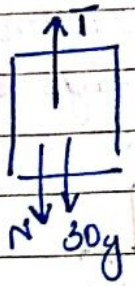
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Q4



Find the force exerted on lift by man (weight of man) as shown by weighing machine.





$T + N = 60g$

$T = N + 30g$

$\Rightarrow T - N = 30g$

$\therefore 2T = 90g$

$\Rightarrow T = 45g$

$\Rightarrow T = 450N$

$N = 600 - 450$

$= 150N$.

Physicswallah

CONSERVATION OF MOMENTUM

In an isolated system, the total momentum remains constant or conserved.

If $F_{net} = 0$ & $F_{net} = \frac{\Delta p}{\Delta t}$

then,

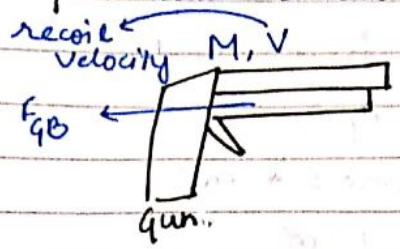
$\Delta p = 0$

change in momentum = 0

so p remains constant (conserved).

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Gun and Bullet System



$F_{net} = 0$

$\vec{F}_{BG} + \vec{F}_{GB} = 0$

P is conserved.

$P_{initial} = P_{final}$

$\Rightarrow 0 = \vec{P}_g + \vec{P}_b$ or $\vec{P}_g = -\vec{P}_b$

Q $M = 5kg$ $m = 0.2kg$

$V = ?$ $v = 200 ms^{-1}$

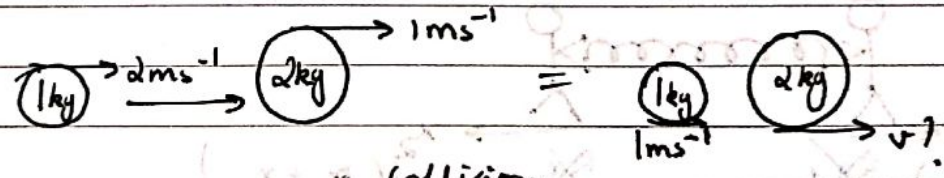
$P_i = P_f$

$\Rightarrow 0 = MV - Mu$

$\Rightarrow 5V - 40 = 0$

$\Rightarrow 5V = 40$

$\Rightarrow V = 8 ms^{-1}$

Q 

Collision

As there is no external force,

$P_i = P_f$

$\Rightarrow 1 \times 2 + 2 \times 1 = 1 \times 1 + 2v$

$\Rightarrow 3 = 2v$

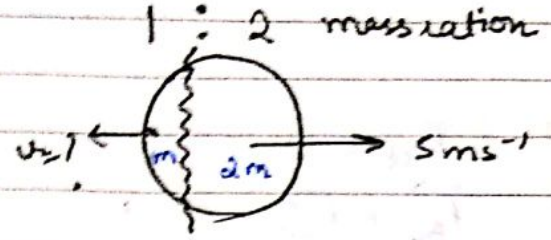
$\Rightarrow v = \frac{3}{2} ms^{-1} = 1.5 ms^{-1}$

Physicswallah

Spiral

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Q



$F_{net} = 0$

so p is conserved.

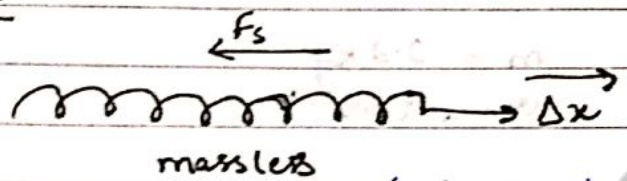
$p_i = p_f$

$\Rightarrow 0 = 2m \times 5 - mv$

$\Rightarrow mv = 10m$

$\Rightarrow v = 10\,ms^{-1}$

SPRING FORCE



SPRING FORCE

variable force (more Δx, more force)

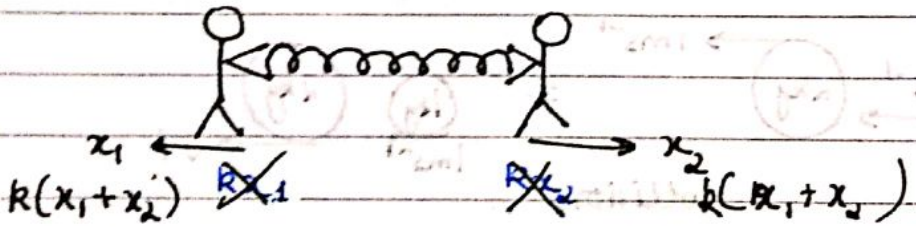
$F_s = -k\Delta x$

spring constant

Hooke's law

direction opposite of Δx

elongation in spring / compression in spring / change in length of spring.

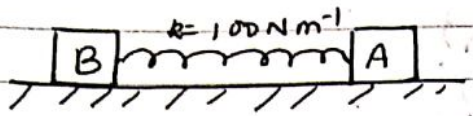


$F_{sp} = k(x_1 + x_2)$

Look at net compression or elongation.

Physicswallah

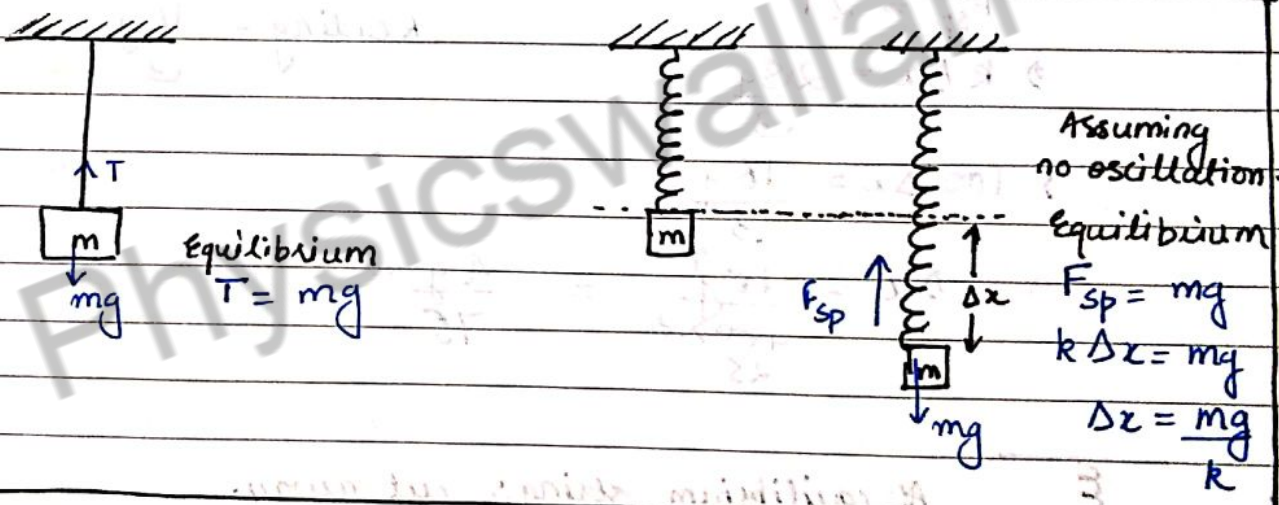
Scanned by CamScanner

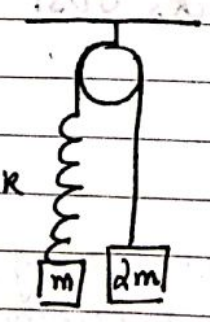
Q 

Find f_s if both blocks are displaced by 1m each
(a) in same direction
(b) in opposite direction

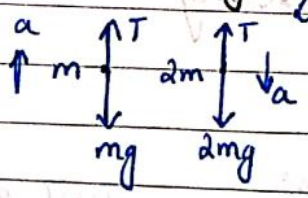
(a) Net change = 0
$f_s = 0$

(b) Net elongation = 1 + 1 = 2m
$f_s = k\Delta x$
$= 100 \times 2$
$= \underline{\underline{200\,N}}$



Q 

Find elongation of spring.
Let us assume a string instead of the spring



$2mg - T = 2ma$
$T - mg = ma$
$\therefore mg = 3ma$
$a = g/3$

$T = mg + \frac{mg}{3} = \frac{4mg}{3}$
$F_{sp} = T = 4mg/3$
$\therefore k\Delta x = \frac{4mg}{3}$
$\Delta x = \frac{4mg}{3k}$

Spiral

Physicswallah

Scanned by CamScanner

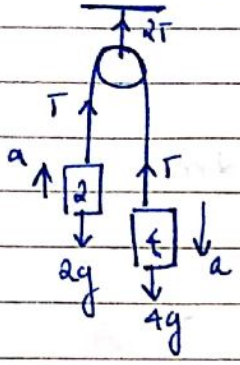
Spring Balance
weighing machine
Weight shown / Reading / = Spring force

Date:
Page No.:

Q Spring Balance, $k = 100\ N m^{-1}$ (Elongation? Reading?)



Assuming a string,



$4g - T = 4a$

$T - 2g = 2a$

$\therefore 2g = 6a$

$\Rightarrow a = \frac{g}{3}$

$F_{sp} = 2T$

$\Rightarrow k\Delta x = 2 \times \frac{8g}{3}$

$\Rightarrow 100\Delta x = \frac{16g}{3}$

$\Rightarrow \Delta x = \frac{16g}{100 \times 3} = \frac{4g}{75}\ m$

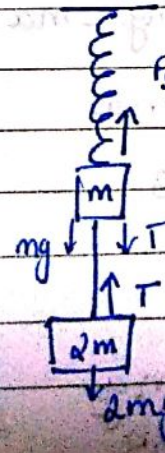
Reading $= \frac{16g}{3}$

Q



At equilibrium, string is cut away.
Find the acceleration of both blocks JUST AFTER cutting the string

At equilibrium,



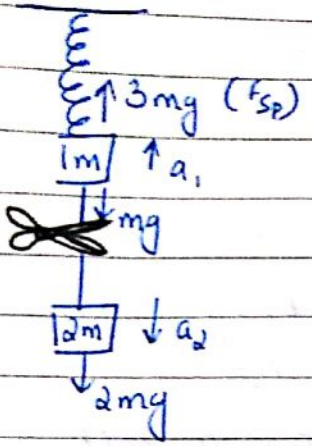
Just after cutting, the tension becomes zero but the spring force remains the same.

Spiral

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$3mg - mg = ma_1$

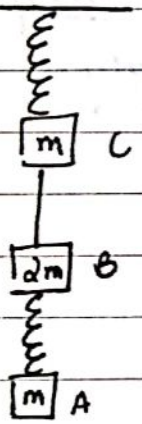
$\Rightarrow 2mg = ma_1$

$\Rightarrow a_1 = 2g \ ms^{-2}$

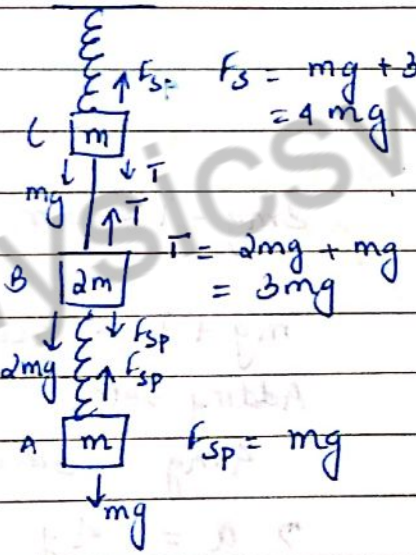
$2mg = ma_2$

$\Rightarrow a_2 = g \ ms^{-2}$

Q After system attains equilibrium, string is cut. Find a_A, a_B, a_C just after cutting.



At equilibrium,



$F_s = mg + 3mg = 4mg$

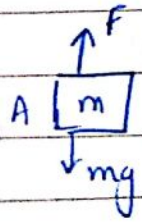
$T = 2mg + mg = 3mg$

$F_{sp} = mg$

Just after cutting,

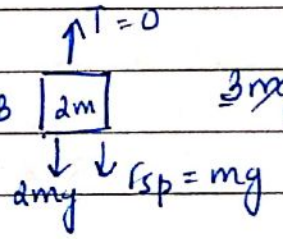
$T = 0$ but

F_s & F_{sp} are same.



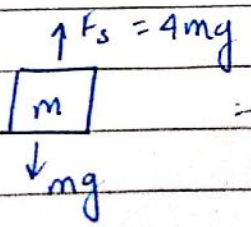
$mg - mg = ma$

$\Rightarrow a_A = 0$



$3mg = 2ma$

$\Rightarrow a_B = \frac{3g}{2}$



$4mg - 3mg = ma$

$\Rightarrow 3mg = ma$

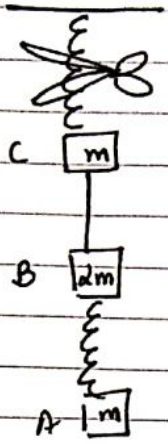
$\Rightarrow a_C = 3g$

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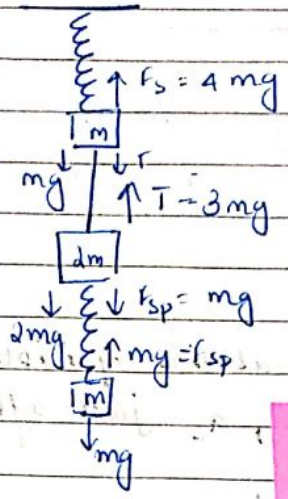
Spiral

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Q

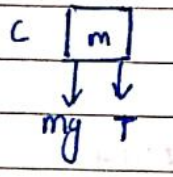


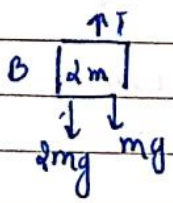
Spring is cut.
Find a_A, a_B & a_C.

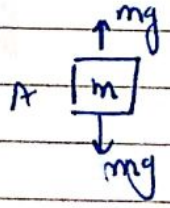


If spring is cut $F_{sp} = 0$
Tension is a self adjusting force and it adjusts itself accordingly.

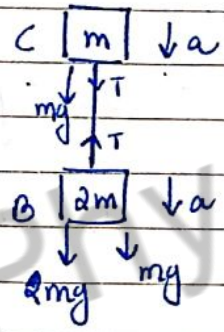
Same string se do mass connected hon to acceleration same hoga. $\vec{a}$







$a_A = 0$



$3mg - T = 2ma$

$mg + T = ma$

Adding both,

$4mg = 3ma$

$\Rightarrow a = \frac{4g}{3}$

$a_A = 0$

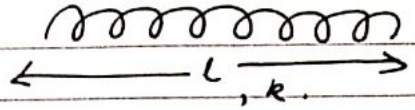
$a_B = \frac{4g}{3}$

$a_C = \frac{4g}{3}$

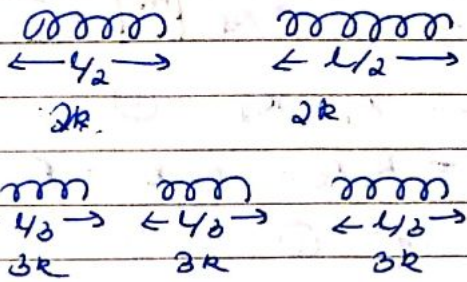
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SPRING CONSTANT 'k'

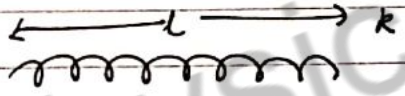


$$k \propto \frac{1}{l}$$

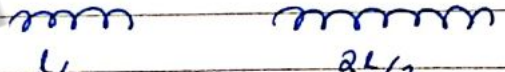


$$k = \frac{C}{l}$$

$\Rightarrow k \times l = \text{constant}$



Eg (i) Ok is cut and ratio of two springs in length is 1:2. Find new spring constant



So, $5k = 3k$ $\quad 5C = \frac{3k}{2}$

(ii) Spring is cut into ratio of 3:5.



$5C = \frac{8k}{3}$ $\quad 5C = \frac{8k}{5}$

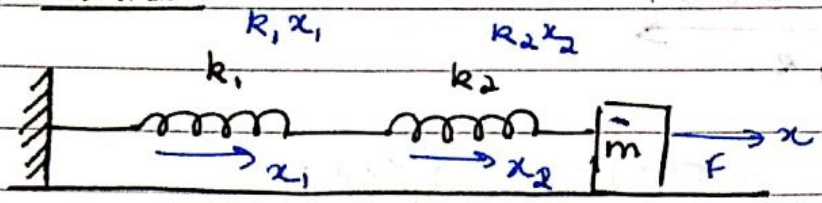
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COMBINATION OF SPRING

(i) SERIES



$$x = x_1 + x_2 \quad \text{— (1)}$$

$$k_1x_1 = k_2x_2 = k_e x = F$$

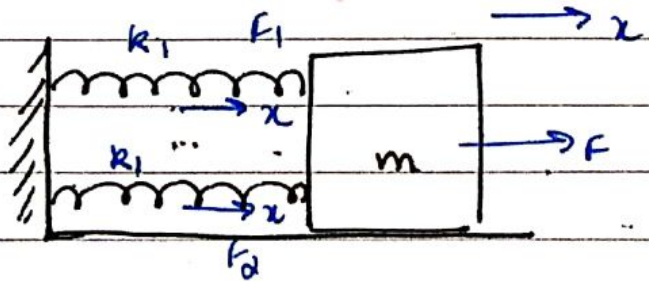
Taking eqⁿ (1)

$$x = x_1 + x_2$$

$$\Rightarrow \frac{F}{k_e} = \frac{F}{k_1} + \frac{F}{k_2}$$

$$\Rightarrow \boxed{\frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2}}$$

(ii) PARALLEL



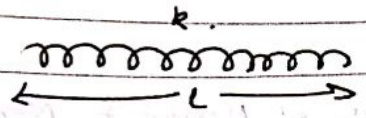
$$F = F_1 + F_2$$

$$\Rightarrow k_e x = k_1 x + k_2 x$$

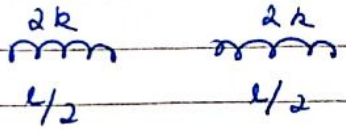
$$\Rightarrow \boxed{k_e = k_1 + k_2}$$

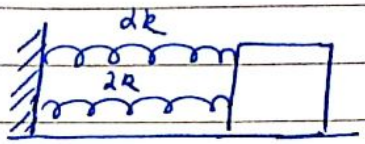
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Q. 

Broken in two parts & those two parts are joined in parallel find new 'k'





$R_e = k_1 + k_2$

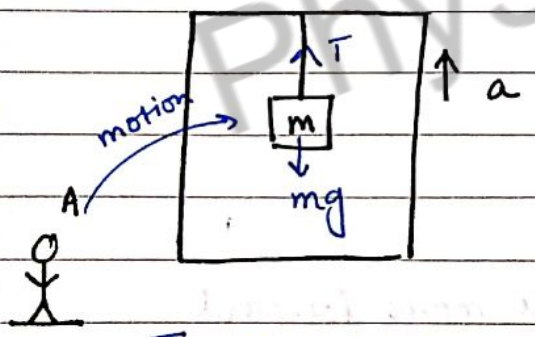
$= 2k + 2k$

$= 4k$

OBSERVER → FRAMES

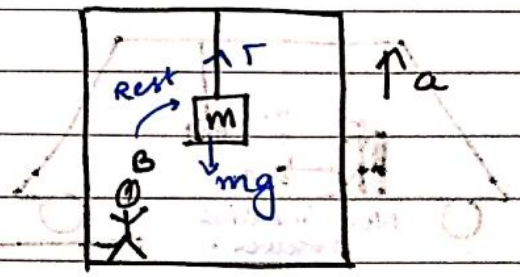
→ Observer $a = 0$, $\vec{v}$ = constant
INERTIAL FRAME
all the laws of Newton are valid

→ Observer $a \neq 0$, $\vec{v}$ = change
Newton's laws of motion are not valid
NON-INERTIAL FRAME



$T - mg = ma$

$\Rightarrow T = m(g + a)$



$T = mg$

This answer is wrong.
The observer B is a non-inertial frame.

Newton's Laws hold only for INERTIAL OBSERVER.
To battle this Pseudo force was introduced.

Physics Wallah

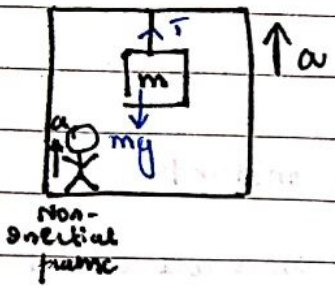
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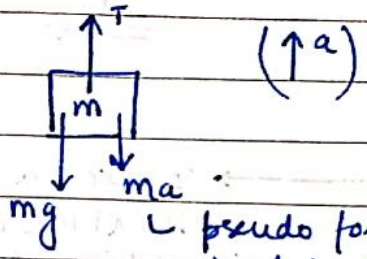
PSEUDO FORCE

Pseudo force is used to apply Newton's Laws of Motion to Non-Inertial frame. It is a fake force.

$$F_p = -ma$$

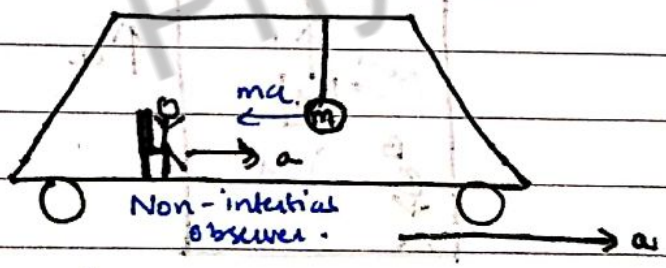
- F_p: to be applied on the object opp. to the direction of acceleration.
- m: mass of object under study
- a: acceleration of observer





↳ pseudo force on the object by the observer opposite to the direction of acceleration.

$$T = m(g + a)$$



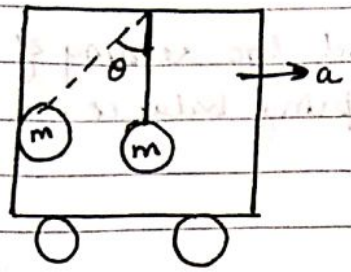
The pendulum moves back when the car moves forward due to the pseudo force of ma opposite to the direction of acceleration on it.

Physicswallah

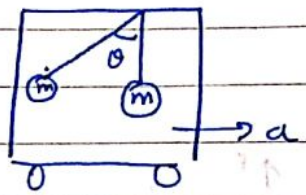
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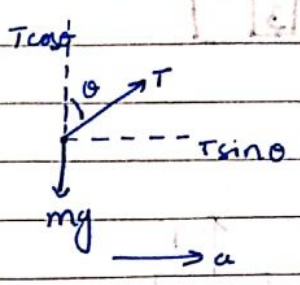
Q



Find θ where m rests in equilibrium.



INERTIAL



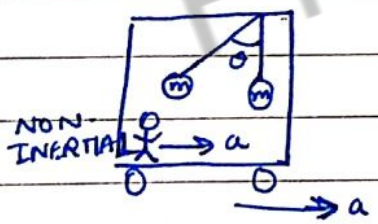
$T\cos\theta = mg$ — (1)

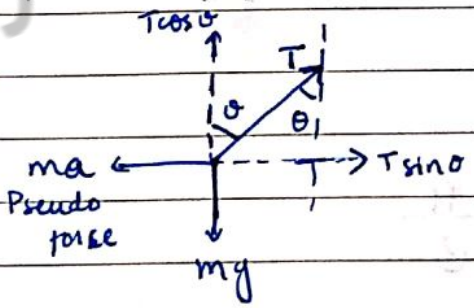
$T\sin\theta = ma$ — (2)

$\frac{(2)}{(1)} \Rightarrow \frac{T\sin\theta}{T\cos\theta} = \frac{ma}{mg}$

$\Rightarrow \tan\theta = \frac{a}{g}$ $\quad \theta = \tan^{-1}\left(\frac{a}{g}\right)$

OR





$T\cos\theta = mg$ — (1)

$T\sin\theta = ma$ — (2)

$\frac{(2)}{(1)} \Rightarrow \frac{T\sin\theta}{T\cos\theta} = \frac{ma}{mg}$

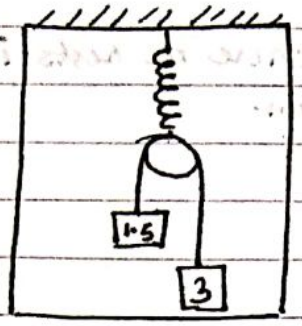
$\Rightarrow \tan\theta = \frac{a}{g}$ $\quad \theta = \tan^{-1}\left(\frac{a}{g}\right)$

Spiral

Physicswallah

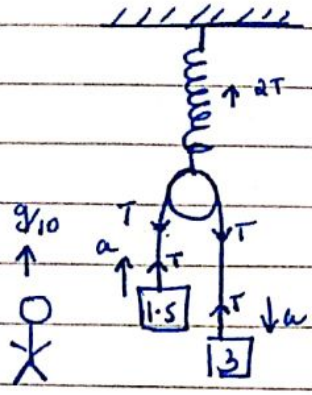
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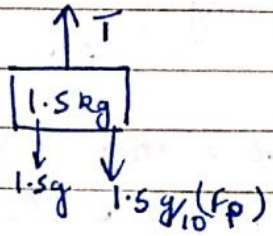
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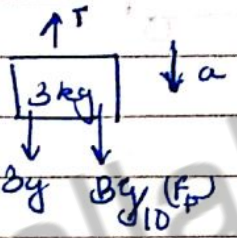


$\uparrow g/10$

Find the reading of spring balance.







$T - 1.5g - 1.5\,g/10 = 1.5a$

$-T + 3g + 3g/10 = 3a$

$1.5g + 1.5\,g/10 = 4.5a$

$\Rightarrow 15 + 1.5 = 4.5a$

$\Rightarrow 16.5 = 4.5a$

$\therefore a = \frac{165}{45} \frac{11}{3} = \frac{11}{3}$

$T = 30 + 3 - 11$

$= 22N$

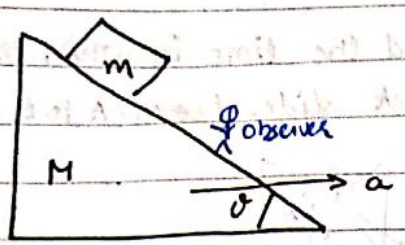
$F_{sp} = 2T$

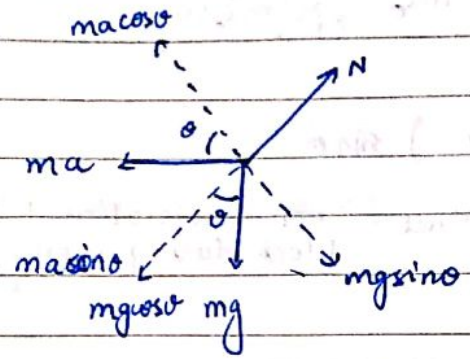
$= 44N$

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Find acceleration of M such that m is at rest wrt M.





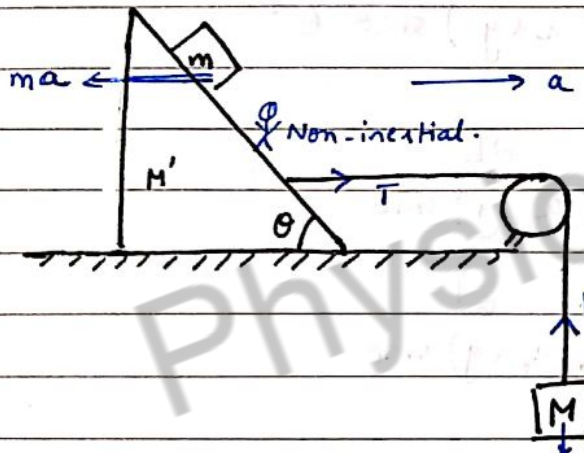
As the mass should be at rest,

$N = ma\sin\theta + mg\cos\theta$

$ma\cos\theta = mg\sin\theta$

$\Rightarrow a = g\frac{\sin\theta}{\cos\theta}$

$\Rightarrow a = g\tan\theta$



What should be mass of hanging block ($M = ?$) for which 'm' does not slide over M'.

The acceleration required for the mass m to stay at rest is $g\tan\theta$

Considering M' & m to be one system,

$T = (M'+m)a$ — (1)

& for mass M

$Mg - T = Ma$ — (2)

Adding (1) & (2),

$Mg = (M'+m)a + Ma$

$\Rightarrow M(g-a) = (M'+m)a$

$\Rightarrow M = \frac{(M'+m)a}{g-a}$

Putting the value of $a = g\tan\theta$

$M = \frac{(M'+m)\tan\theta}{1-\tan\theta}$

or

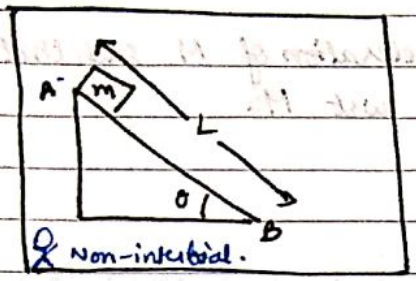
$M = \frac{(M'+m)}{\cot\theta - 1}$

Spiral

Physicswallah

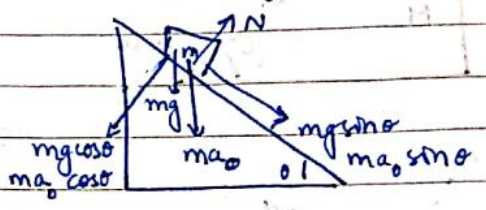
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Q



$\uparrow a_0$

Find the time in which the block slides from A to B.



Force which causes slipping = $(ma_0 + mg)\sin\theta$

& $(ma_0 + mg)\sin\theta = ma_{net}$ (a_{net} = net acceleration of block during slipping)

$\Rightarrow (a_0 + g)\sin\theta = a_{net}$.

$u = 0\ ms^{-1}$

$s = L$

$a = (a_0 + g)\sin\theta$

$s = ut + \frac{1}{2}at^2$

$\Rightarrow L = \frac{1}{2}(a_0 + g)\sin\theta\, t^2$

$\Rightarrow t^2 = \frac{2L}{(a_0 + g)\sin\theta}$

$\Rightarrow t = \sqrt{\frac{2L}{(a_0 + g)\sin\theta}}$

FRICTION

Friction is the force which opposes the relative motion of two bodies (or tendency of relative motion).

Friction force acts tangentially along the contact.

STATIC
when tendency of relative motion is there (rest)

KINETIC
when relative motion is actually there (motion)

Spiral

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KINETIC FRICTION

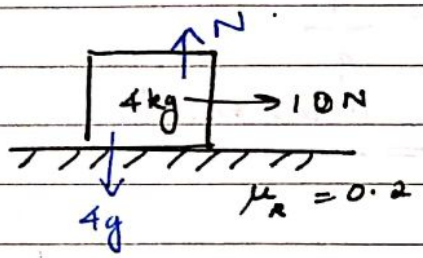
* Relative motion.
* Constant friction force
  f_R = constant.
* $f_R \propto N$

$$\boxed{f_R = \mu_R N}$$

μ_R → coefficient of kinetic friction. (depends on surface which are in contact (determined experimentally))

* direction is opposite to relative motion.

Ex.



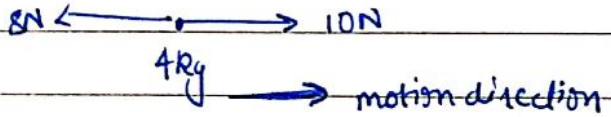
① Find friction force
② $\vec{a}$ of 4kg block.

$N = 4g = 40N$

$f_R = \mu_R N$

$= 0.2 \times 40$

$= 8N$ (opposite the direction of motion)

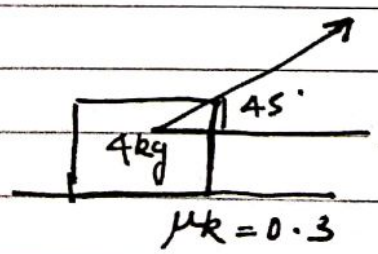


$\therefore \quad 10 - 8 = 4a$

$\Rightarrow 2 = 4a$

$\Rightarrow a = 0.5 ms^{-2}$

Ex.

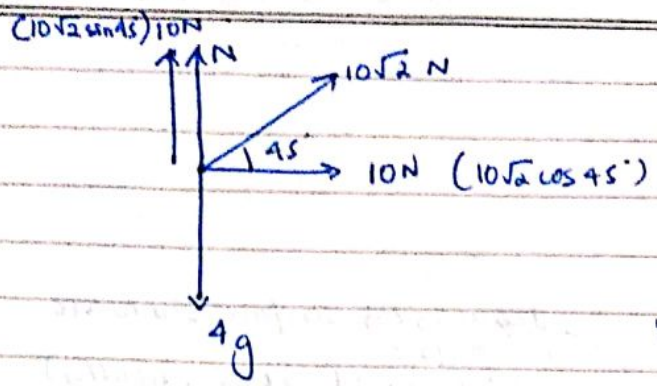


① Find N
② Find f_k
③ find a

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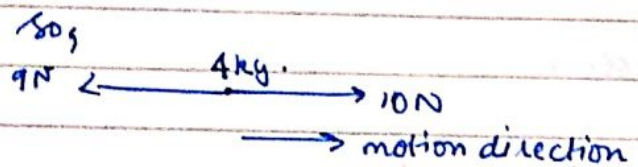


As the block is in vertical equilibrium

$N + 10 = 4g$

$\Rightarrow N = 40 - 10 = 30N$.

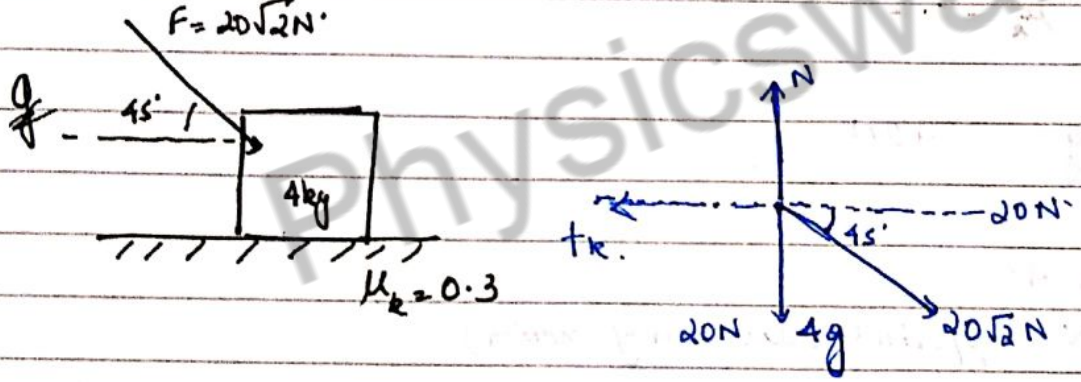
$f_R = \mu_R N = 0.3 \times 30 = 9N$.



$\therefore 10 - 9 = 4 \times a$

$\Rightarrow a = \frac{1}{4}$

$\Rightarrow a = 0.25\ ms^{-2}$



$N = 20N + 40N$

$\Rightarrow N = 60N.$

$f_k = 0.3 \times 60$

$= 18\ N$

$a = \frac{20 - 18}{4} = 0.5 ms^{-2}$

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$f_{limiting} > f_{kinetic}$.
(just about to move)

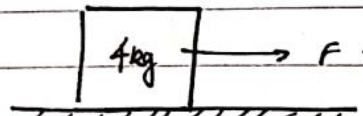
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STATIC FRICTION

* Tendency of relative motion (bodies at rest)
* variable friction → self adjusting
* $0 \leq f_s \leq f_{limiting}$

$$\boxed{f_{lim} = \mu_s N}$$ μ_s → coefficient of static friction.

Body moves only if $F_{applied} >$ Maximum static friction or f_{lim}

Q 

$\mu_s = 0.2$.

Find friction & $\vec{a}$:

$N = 40N$.

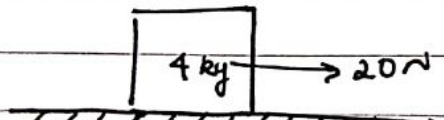
If body is at rest, Friction force = Applied force.

$f_{lim} = \mu_s N$
$= 0.2 \times 40$
$= 8N$.

As $f_{applied} < f_{lim}$, the body will not move.

$\vec{a} = 0$

Friction = 6N (opposite to applied force)

Q 

$\mu_s = 0.6$ $\mu_k = 0.4$

Find friction force and acceleration.

$f_{lim} = \mu_s N$ $(N = 40N)$
$= 0.6 \times 40$
$= 24N$

As $f_{applied} < f_{lim}$, the body will not move.

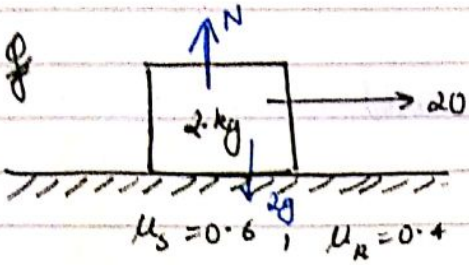
Friction force = 20N

$\vec{a} = 0$

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Find friction and $\vec{a}$.

$N = 2g = 20\ N$

$f_{lim} = \mu_s N = 0.6 \times 20$

$= 12\ N$

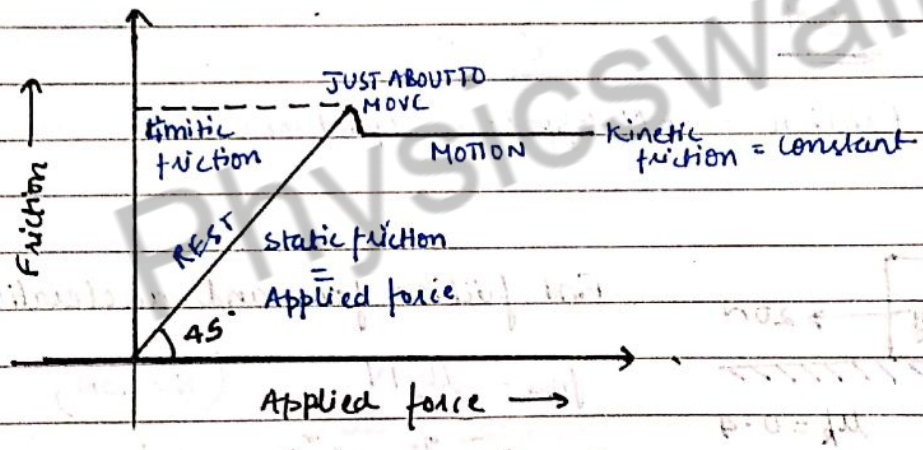
As $f_{app} > f_{lim}$, body will move.

$f_k = \mu_k N = 0.4 \times 20$

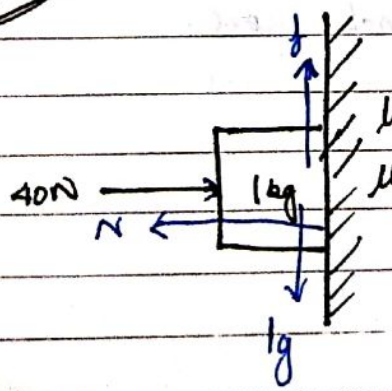
$= 8\ N$

$a = \frac{20 - 8}{2} = \frac{12}{2} = 6\ ms^{-2}$

GRAPH OF FRICTION



IMP



So Friction = 10N

Find force of friction.

In this case, since there is no horizontal motion, $N = 40N$

Also the motion of block in in downwards direction and friction opposes motion so it opposes the downward slipping.

$f_{lim} = \mu_s N = 0.3 \times 40 = 12\ N$

As the downward force is 10N and is less than limiting friction, the block will not move.

Spiral

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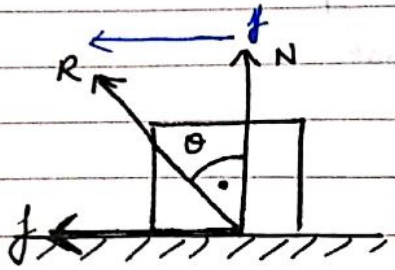
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ANGLE OF FRICTION

During contact between two surfaces, the contact forces are two.

1. Normal Reaction
2. Frictional force (incase of motion or tendency to move)

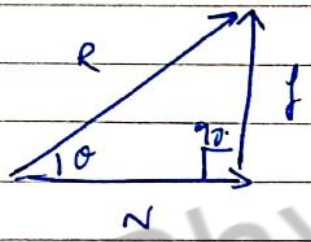
Let R be the RESULTANT CONTACT FORCE then,



If the object is just about to move,

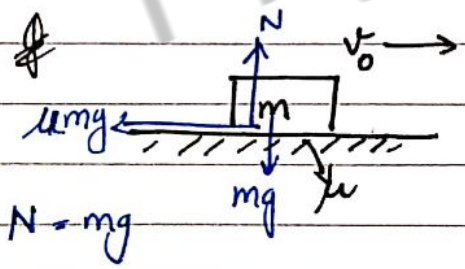
Frictional force = $f_L = \mu N$.

In this case, the angle formed by the resultant contact force and Normal Reaction is the angle of friction.



$$\tan\theta = \frac{f}{N} = \frac{\mu N}{N}$$

$$\boxed{\tan\theta = \mu}$$



Find the time after which it stops and also calculate displacement of block.

$$f_k = \mu N$$
$$= \mu mg$$

$$a = \frac{F}{m} = \frac{\mu mg}{m} = -\mu g$$

$u = v_0$

$v = 0$

$t = t$

$v = u + at$

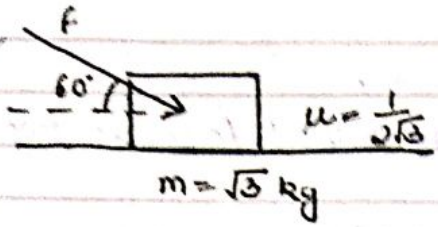
$\Rightarrow 0 = v_0 - \mu g t$

$\Rightarrow t = \frac{v_0}{\mu g}$

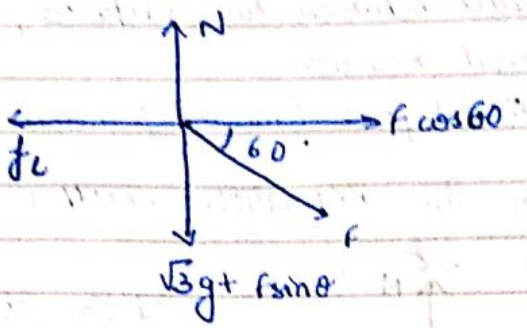
Spiral

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Q. 

Find maximum value of F such that block doesnot move.



$N = \sqrt{3}g + F\sin\theta$

$\Rightarrow N = 10\sqrt{3} + F\sin 60°$

For the block to not move,

$F\cos 60° = f_L$ & $f_L = \mu N$

So, $F\cos 60° = \mu N$

$\Rightarrow F\cos 60° = \frac{1}{2\sqrt{3}}(10\sqrt{3} + F\sin 60°)$

$\Rightarrow F\cos 60° = 5 + \frac{F}{4}$

$\Rightarrow \frac{F}{2} - \frac{F}{4} = 5 \quad \Rightarrow \frac{2F - F}{4} = 5$

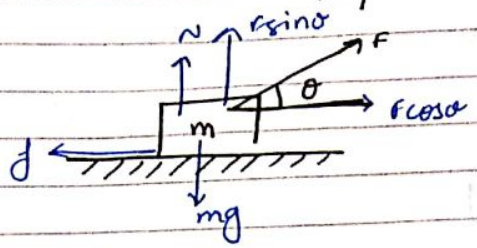
$\therefore \underline{F = 20\ N}$

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Q. Which is easier, push or pull, when force is at an angle?

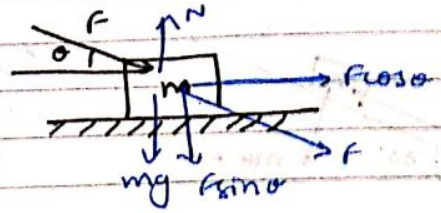


$N + F\sin\theta = mg$

$N = mg - F\sin\theta$

$f = \mu N$

$$f = \mu(mg - F\sin\theta)$$



$N = mg + F\sin\theta$

$f = \mu N$

$$f = \mu(mg + F\sin\theta)$$

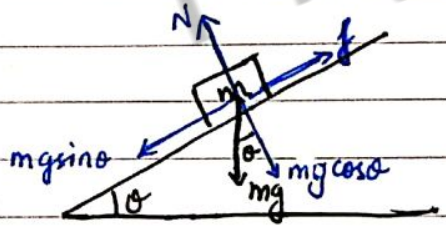
As friction is more in the push case, it is difficult to push.

ANGLE OF REPOSE

Angle of repose is the maximum angle of the incline at which the body stays at rest

OR

It is the maximum angle at which block is just about to slide.



$N = mg\cos\theta$

$f = \mu N$

$= \mu mg\cos\theta$

Value at which block will stay at rest,

$f = mg\sin\theta$

$\Rightarrow \mu mg\cos\theta = mg\sin\theta$

$\Rightarrow \tan\theta = \mu$

$$\tan\theta_R = \mu$$

Spiral

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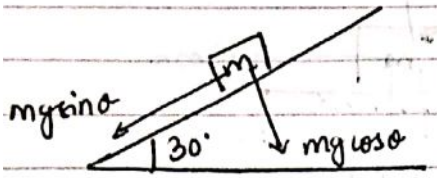
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Given $\theta < \theta_R$
No sliding
Body at rest
static friction = Applied force

$\theta = \theta_R$
Just about to slide.
Limiting friction $f_L = \mu_s N$

θ_R
ing starts
otion.
Friction $f_k = \mu_k N$

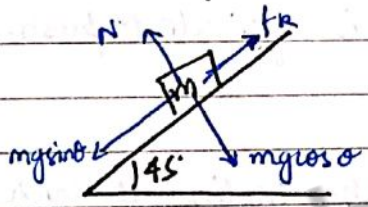
Q If angle of repose is 45°, find whether body is at rest or motion. Calculate friction and $\vec{a}$.



$\theta_R = 45°$
$30° < \theta_R$
No sliding.

static friction = applied force
$= mg\sin\theta$

Q If angle of repose is 37°, Find if body is at rest or motion. Calculate f.

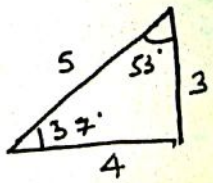


$\theta_R = 37°$
$45° > \theta_R$
Body will slide.

$mg\sin 45° - f_R = ma$
$\Rightarrow mg\sin 45° - \mu N = ma$
$\Rightarrow mg\sin 45° - \mu mg\cos 45° = ma$
$\Rightarrow \frac{1}{\sqrt{2}} mg(1-\mu) = ma$
$\Rightarrow a = \frac{1}{\sqrt{2}}(1-\mu)$

$f = \mu N$
$= \mu mg\cos 45°$
$= \frac{1}{\sqrt{2}} \mu mg$

μ_s	θ_R	$\tan\theta_R = \mu_s$
1	45°	
0.75 $\left(\frac{3}{4}\right)$	37°	
1.33 $\left(\frac{4}{3}\right)$	53°	



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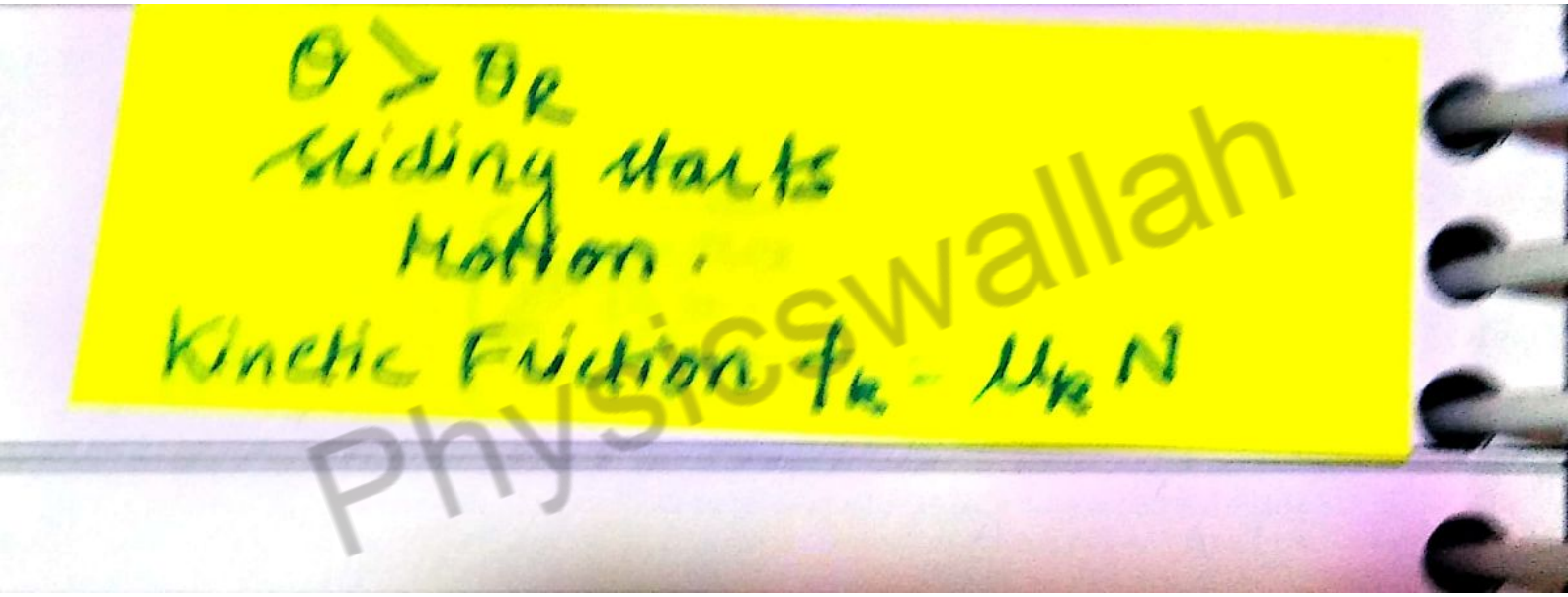
$\theta = \theta_R$

Just about to slide.

Limiting friction $f_L = \mu_s N$

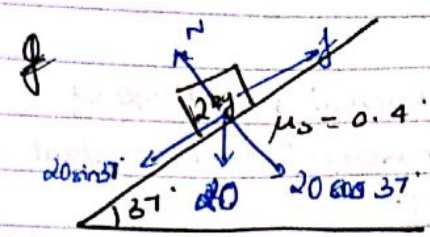
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$\mu_s = 0.4$

$N = 20 \cos 37°$

$= 20 \times \frac{4}{5} = 16\ N$

Limiting friction $= 0.4 \times 16$

$= 6.4\ N$.

Force which causes the mass to slip $= 20 \sin 37°$

$= 20 \times \frac{3}{5} = 12\ N$.

As the applied force > limiting friction, the mass will move.

∴ Frictional force $= f_k$

$= \mu_R N$

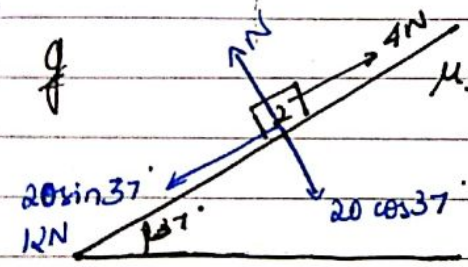
$= 0.3 \times 16$

$= 4.8\ N$

$ma_{net} = 12 - 4.8\ N$

$\Rightarrow a = \frac{7.4}{2}$

$\Rightarrow a = 3.7\ ms^{-2}$



Find f and $\vec{a}$.

Net force causing block to slip $= 12 - 4$

$= 8\ N$

$f_L = 0.4 \times 16$

$= 6.4\ N$.

The object is in motion.

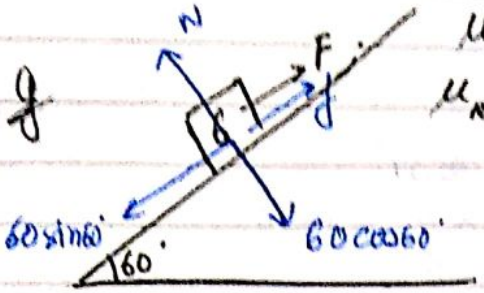
So, $f_k = 4.8\ N$.

$a = \frac{3.2}{2} = 1.6\ ms^{-2}$

Spiral

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$\mu_s = 0.6$

$\mu_k = 0.1$

$F = ?$

for (a) block remains stationary

(b) move downwards with constant velocity.

(c) move upwards with acceleration of $1 ms^{-2}$.

$N = 60 \cos 60°$

$= 60 \times \frac{1}{2} = 30 N$

$f_{lim} = \mu_s N$

$= 0.6 \times 30$

$= 18 N$

Applied force $= 60 \sin 60° - F$ or $F - 60 \sin 60°$

(a) For the block to remain stationary,

$60 \sin 60° - F = f_{lim}$

$\Rightarrow \frac{60\sqrt{3}}{2} - F = 18$

$\Rightarrow F = 30\sqrt{3} - 18$

$\Rightarrow F = 6\sqrt{3}(5 - \sqrt{3})$ N

$\Rightarrow F = 10.392 \times 3.268$

$\Rightarrow F = 35.072176$

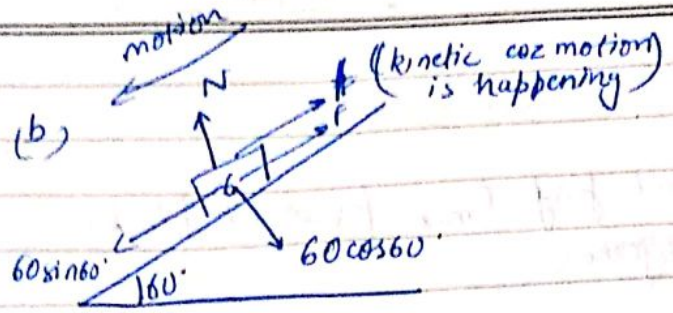
$\Rightarrow F = 35$ N (approx)

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(b)



$f_R = \mu_R N$

$= 0.4 \times 60 \times \frac{1}{2}$

$= 12\ N$

force which slides $= 60 \sin 60°$

$= 52\ N$ (approx)

for constant velocity, $a = 0$.

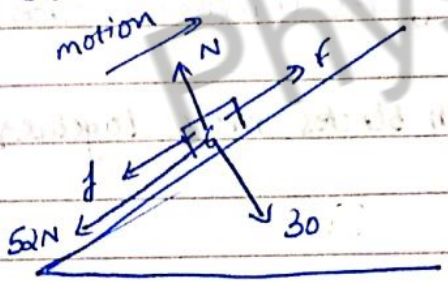
so,

$60 \sin 60° - f_R - F = ma$

$\Rightarrow 52 - 12 - F = 0$

$\Rightarrow F = 52 - 12 = \underline{40\ N}$

(c) Since motion is happening, friction will be kinetic.

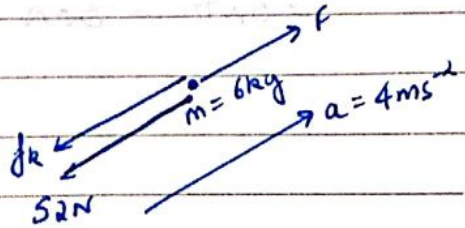


$N = 30\ N$

$f_k = \mu_R N$

$= 0.4 \times 30$

$= 12\ N$.



$\therefore F - (f_R + 52) = ma$

$\Rightarrow F = 6 \times 4 + 12 + 52$

$\Rightarrow F = 24 + 12 + 52$

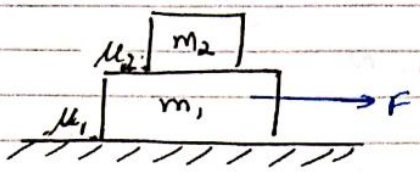
$\Rightarrow F = \underline{88\ N}$

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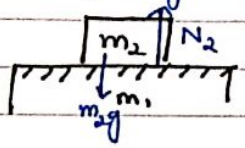
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BLOCK ON BLOCK DIAGRAM

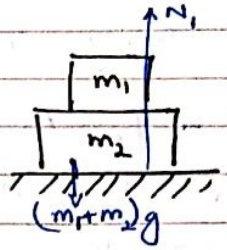
We will find f_{max} for which blocks move together



1. Find $f_{limiting}$ on both surfaces.



$N_2 = m_2 g$

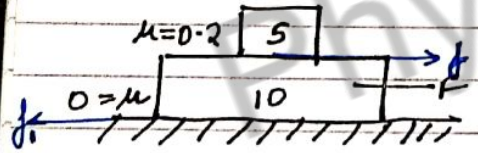


$N_1 = (m_1+m_2)g$

2. Search that block which only has frictional force. Find a_{max} of this block.

3. Use this a_{max} in other block and find F_{max} for together motion.

Q. Find the maximum value of F such the both blocks move together



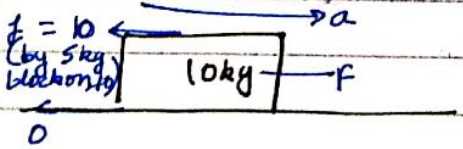
$f_{1\,lim} = 0$

$f_{lim} = \mu N$ $\quad N = m_2 g$

$= 0.2 \times 50$ $\quad = 5 \times 10 = 50N$

$= 10N$

So, a_{max} of 5kg block $= \frac{10}{5} = 2 ms^{-2}$



$f - f = ma$

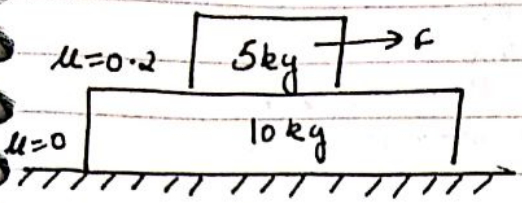
$\Rightarrow F - 10 = 10 \times 2$

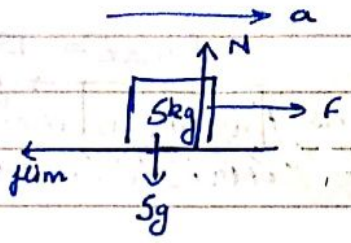
$\Rightarrow F_{max} = 30 N$

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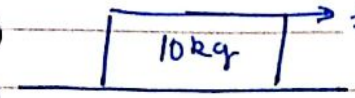
Q find F_{max} so that both blocks move together.





$N = 5 \times 10$
$= 60\ N.$

$f_{im} = \mu N$
$= 0.2 \times 50$
$= 10\ N$



$f_{im} = 10\ N.$

$a_{max} = \frac{10}{10} = 1\ ms^{-2}.$

So, for the upper block,

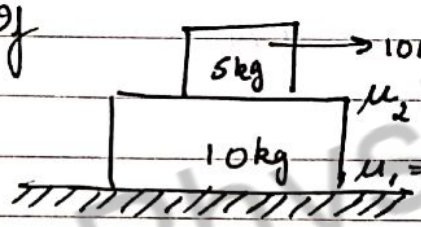
$F - f_{im} = ma$

$\Rightarrow F_{max} = 5 \times 1 + 10 = 15\ N$

$F_{max} = 15\ N$

Blocks move together ⇓ a is same ⇓ No relative motion ⇓ Static friction (f)

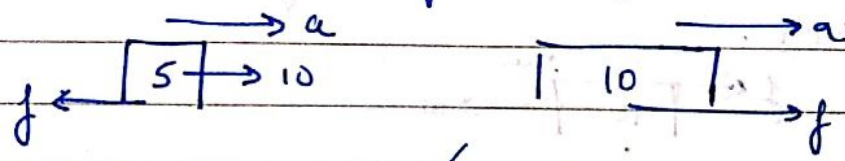
Q



Find a of both blocks

As the force applied is less than f_{max}, both blo move together and acceleration will be the same.

Let the static friction b/w blocks be f.



$10 - f = 5a$

$f = 10a$

$10 = 15a$

$\Rightarrow a = \frac{10}{15} = \frac{2}{3}\ ms^{-2}$

Spiral

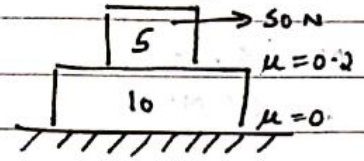
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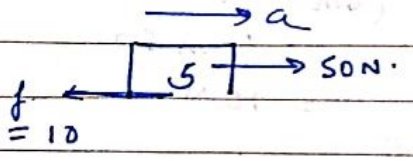
Both move seperately ⇓ $a \neq 0$ ⇓ Relative motion ⇓ Kinetic friction $f_k = \mu_k N$

Q Find a of both blocks & friction between them.

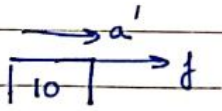


$f_{max} = 15N$.

As $F_{applied}$ is more than f_{max}, the blocks will not move together

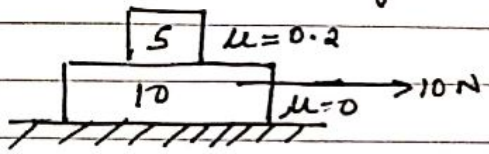


$$a = \frac{50-10}{5} = \underline{8\ ms^{-2}}$$



$$a' = \frac{10}{10} = 1\ ms^{-2}$$

Q Find acceleration of both blocks and friction b/w them.

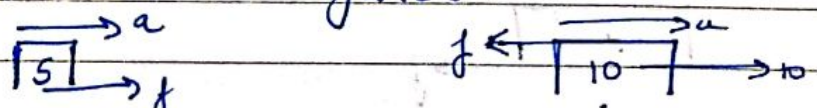


fric b/w blocks $= \mu N$
$= 0.2 \times 50$
$= 10N$

a_{max} for 5 kg block $= \frac{10}{5} = 2\ ms^{-2}$

f_{max} for 10 kg block $= 10 \times 2 + 10$
$= \underline{30\ N}$

As the force applied is less than the maximum force, the blocks will move together.



$10 - f = 10a$
$f = 5a$
$10 = 15a$
$\therefore a = \frac{2}{3}\ ms^{-2}$

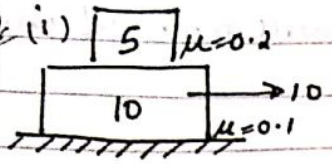
Force b/w blocks $= \frac{10}{3}\ N$

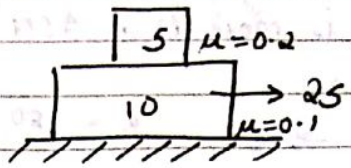
Spiral

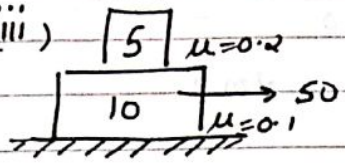
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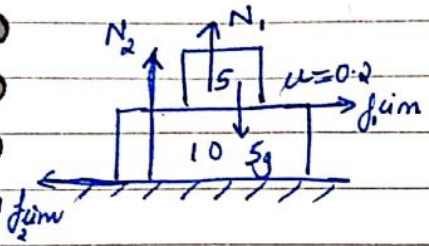
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Q (i) 

(ii) 

(iii) 

Find a of both blocks.

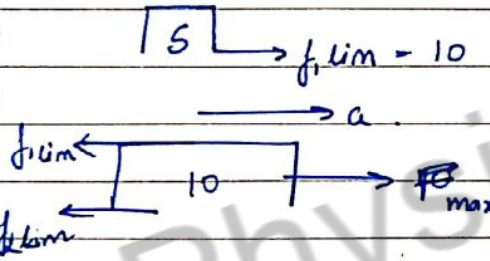


$N_1 = 5g = 50N$

$f_{lim} = 0.2 \times 50 = 10N$

$N_2 = 150\ N$

$f_{lim} = 150 \times 0.1 = 15N$



$a_{max} = \frac{10}{5} = 2ms^{-1}$

$f_{max} - 10 - 15 = m \times a$

$\Rightarrow f_{max} = 10 \times 2 + 10 + 15$

$\Rightarrow f_{max} = 45\ N$

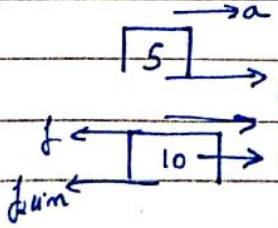
As there is a limiting friction b/w ground & block this time,

$15 \leq F \leq 45$ Both move together

(i) As the force applied is less than the limiting friction, the blocks are at rest. a of block = 0,

(ii) As the force is less than 45N, the blocks will move together.

Let the static friction be f.



$f = 5a$

$25 - 10 - f = 10a$

$15 = 15a$

$\therefore a = 2ms^{-1}$

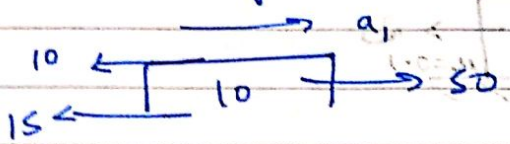
$f = \frac{10}{3}\ N$

Spiral

Physicswallah

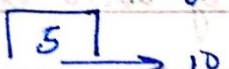
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(iii) As the force is more than 45N, the blocks will move seperately.



$$a_1 = \frac{50 - 10 - 15}{10}$$

$$= \frac{25}{10} = 2.5\,ms^{-2}$$



$$a_2 = \frac{10}{5} = 2\,m\,s^{-2}$$

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